

Formative Assessment 7: Part 2

Applying Normal Distribution on Campus

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Background

- The **objective** of this study is to measure and analyze Far Eastern University's facility utilization data using the occupancy proportion $\left(\frac{Occupied}{Capacity}\right)$.
- **Research Question:** Does campus facility occupancy rate $\left(\frac{Occupied}{Capacity}\right)$ follow a normal distribution, and what do its distribution characteristics reveal about campus space utilization?

- The variable being studied is that of the occupancy proportion $\left(\frac{Occupied}{Capacity} \right)$.
- The gathered sample includes observation data from the FEU Pavilion from 10:00 to 10:30 and the second and third floors of the FEU Library from 12:00 to 1:00. The sample size consists of 143 instances of varying capacities ranging from a maximum of 12 tables to a minimum of 2 or 3.

Frequency Distribution Table

The table below displays the frequency distribution for the calculated proportions across 143 observations.

Interval	Count
0.01 - 0.20	2
0.21 - 0.40	21
0.41 - 0.60	42
0.61 - 0.80	35
0.81 - 1.00	43

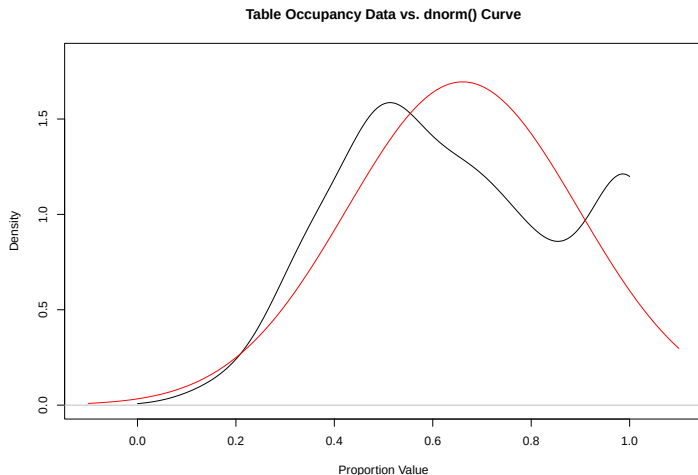
Mean and Standard Deviation

The mean and standard deviation are shown below:

- **Mean:** 0.6609
- **Standard Deviation:** 0.2354

Normal Distribution Construction

The plot below displays the empirical distribution curve overlaid with the normal distribution curve.



Percentage of Data Within 1, 2, and 3 Standard Deviations from the Mean

Based on the occupancy proportion data gathered from the campus facilities, the actual distribution spreads across standard deviation thresholds as follows:

- **Within $\pm 1\sigma$ ($\mu \pm \sigma$):** 59.4% (*Theoretical: 68.3%*)
- **Within $\pm 2\sigma$ ($\mu \pm 2\sigma$):** 98.6% (*Theoretical: 95.4%*)
- **Within $\pm 3\sigma$ ($\mu \pm 3\sigma$):** 100% (*Theoretical: 99.7%*)

These empirical values can show how close the campus occupancy rate follows a theoretical normal distribution.

Skewness, Outliers, and Shape

The distribution is approximately symmetric (skewness = 0.1048), has 0 outliers (bounds = -0.0312 to 1.3854), and its shape suggests that occupancy proportions are relatively balanced around the mean but are not perfectly normally distributed.

Findings

- The data shows that campus occupancy is generally moderate to high, with an approximate symmetric distribution but its not close to a normal distribution.
- These findings demonstrate that Pavilion and Library utilization varies across observations; thus, scheduling and space allocation can be adjusted for periods of higher occupancy (such as exam seasons) or lower occupancy (such as synchronous learning days).
- The data can help the university monitor facility utilization and make informed decisions about campus space management.
- To optimize campus space, the university should implement real-time occupancy tracking and dynamic scheduling that shifts resources during peak exam seasons and utilizes low-occupancy synchronous learning days for maintenance.

Summary

- **Sample size:** 143
- **Mean (μ):** 0.6609
- **Standard deviation (σ):** 0.2354
- **Skewness:** 0.1048
- Q_1 : 0.5
- Q_3 : 0.8542
- **IQR:** 0.3542
- **Outlier Bounds:** -0.0312 to 1.3854
- **Number of Outliers:** 0
- **Within $\pm 1\sigma$:** 59.4%
- **Within $\pm 2\sigma$:** 98.6%
- **Within $\pm 3\sigma$:** 100%
- Analysis codes can be found in the .Rmd file!